

The crypto world trades at tea time. Intraday evidence from centralized exchanges across the globe

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Abstract

We study intraday patterns of returns, trading activity, volatility and illiquidity in cryptocurrency markets. We analyze intraday data from 1,940 trading pairs traded on 38 centralized exchanges around the globe and show that there are pronounced intraday patterns. These patterns are similar across the variables under consideration, across trading venues, and across currency pairs. Characteristics of the trading venues (such as their location) and the traded currency pairs explain some, but not all of the commonality in intraday patterns.

Keywords: cryptocurrencies, seasonalities, time of the day patterns, liquidity

JEL: G12, G14

1. Introduction

It is a well documented stylized fact that asset returns, trading activity, volatility, and liquidity follow specific intraday patterns. There are at least three reasons for these patterns to emerge. First, there are institutional features such as the overnight trading halt that cause specific intraday patterns. Private and public information that has accumulated overnight is incorporated into prices after market opening, resulting in higher trading activity, higher volatility, and

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lower liquidity. Similarly, the desire of traders to close positions before the overnight trading halt leads to higher trading volume before and at the market close. Second, the flow of public and private information relevant to the value of an asset will affect intraday patterns. For example, trading activity and volatility will be higher at times of the day with more news events. Third, trader behavior may affect intraday patterns, possibly leading to regularities such as lower trading activity during lunchtime.

Cryptocurrency markets provide an ideal test bed to analyze intraday patterns for several reasons. These markets are open 24 hours, seven days a week. Thus, there are no regular trading halts that affect intraday patterns. Second, transactions are not settled on a daily basis (as is the case in equity markets) but are rather settled immediately. Consequently, there is no pressure to trade at a specific time of the day to avoid holding positions for an extended period of time (e.g., overnight). Moreover, the same currency pairs (either cryptocurrency against cryptocurrency or cryptocurrency against fiat currency) are traded on a large number of exchanges on different continents and in different time zones. While the flow of information is the same across all trading venues, trader population and trader behavior are venue-specific. This makes it possible to decouple the effects on intraday patterns of trader behavior from those of information flow. In this paper, we leverage these characteristics of cryptocurrency markets. Specifically, we use an extensive data set covering 1,940 trading pairs¹ traded on 38 cryptocurrency exchanges around the globe.

Our analysis proceeds as follows. In a first step, we regress, separately for each trading pair, the variable of interest on a set of 24-hour-of-day dummies. The coefficients for these dummy variables are the basis for our analysis. To describe the intraday patterns, we show figures for the overall average and the continent-specific averages of these demeaned coefficients. We then calculate the correlations between the coefficients for all combinations of trading pairs in our

¹As described in section 2, a trading pair is defined at the currency pair - exchange level, implying that the same currency pair is included in our data several times if it is traded on several venues.

sample. We use this pairwise setting to capture similarities between intraday patterns both across currency pairs (i.e., different currency pairs traded on the same trading venue) and within currency pairs (i.e., the same currency pair traded on different trading venues). We further use hierarchical cluster analysis
40 to identify groups of trading pairs with similar intraday patterns. Finally, we estimate regressions where pairwise correlations are regressed on variables that capture characteristics of the currency pair in question (e.g., does it contain a fiat currency?) and the venues where they are traded (e.g., are the venues located on the same continent?). The analysis enables us to identify characteristics that
45 affect the similarity of intraday patterns across trading venues and currency pairs.

There is not much theory to inform our empirical analysis. In their seminal paper on intraday patterns, Admati and Pfleiderer (1988) develop a theoretical model in which uninformed liquidity traders and informed insiders trade a risky
50 asset. They show that liquidity traders have an incentive to pool their trades, i.e., to trade at the same time. This pooling of trades leads to a concentration of trading in particular periods during the day. The model further implies that volatility is higher during periods of concentrated trading. These patterns yield the predictions that there are periods of increased trading activities, and that
55 volatility also peaks at these periods of concentrated trading. The logic of the Admati and Pfleiderer (1988) model extends to a multi-market setting. If the same asset is traded on different venues, liquidity traders will have an incentive to concentrate their trading on all venues at the same time. This, in turn, implies that the intraday patterns will be correlated across venues. To the extent that
60 information arrival is correlated across currency pairs, we also expect correlated intraday patterns across currency pairs.

Our results reveal that there are pronounced intraday patterns. Returns are lowest in the early morning hours (in Coordinated Universal Time, UTC) and highest in the early afternoon and evening. Trading volume, volatility
65 and illiquidity are lowest in the early morning hours and in the evening and highest in the afternoon and around midnight. When we sort the trading pairs

in our sample by venue location (Americas, Asia, Europe) we find that these patterns (again in UTC, not in local time) are strikingly similar, implying that the patterns are not primarily caused by local factors such as trader population
70 or trader behavior. The correlations are higher for trading activity, volatility and liquidity than for returns. This is a surprising finding, because different patterns of returns can trigger arbitrage activity while different patterns in the other variables do not. Our regression analysis reveals that the intraday patterns are more highly correlated when the trading venues are on the same continent,
75 when the time difference between the venue locations is lower, and when the currencies in the trading pair are similar (e.g. because they share a common currency, such as BTC-ETH and BTC-XRP). Thus, and notwithstanding the high degree of commonality, local factors do play a role in shaping the intraday patterns.

80 Our research is related to several strands of the literature. First, it is related to previous papers that document intraday patterns in cryptocurrency markets (e.g. Baur et al. (2019), Dyhrberg et al. (2018), Eross et al. (2019), Petukhina et al. (2021), Shams (2020), Su et al. (2022)).² Our paper adds to this literature by considering a much larger sample in terms of the number of currency pairs
85 and trading venues we analyze, as well as the number of variables included in our analysis. Second, our research is related to studies of intraday patterns in financial markets more generally. Most of that literature relates to equity markets (e.g. the classical studies by Wood et al. (1985) and McInish and Wood (1992) or more recently Heston et al. (2010)). As mentioned earlier, the cryp-

²There is also a growing literature on calendar and day-of-the-week effects in cryptocurrencies (e.g. Aharon and Qadan (2019), Caporale and Plastun (2019), Dorfleitner and Lung (2018), Kaiser (2019), Kinatader and Papavassiliou (2021), Long et al. (2020), Ma and Tanizaki (2019), Qadan et al. (2021)). Moreover, several papers deal with dynamic market linkages across cryptocurrencies. For example, Hu et al. (2019), Aslanidis et al. (2021) and Bouri et al. (2021) examine the correlation of return and volatility correlations between several cryptocurrencies and find that there are strong and increasing market linkages for both variables over time. Hasan et al. (2022) and Tripathi et al. (2021) focus on commonality in liquidity and document a significant presence of linkages in cryptocurrency markets. However, none of these papers uses intraday observations to examine cryptocurrency linkages or identifies potential determinants over the course of a day.

90 to currency markets we analyze differ from equity markets in that they operate
24 hours a day, and because the same currency pairs are traded on various trad-
ing venues located in different time zones and continents. In this respect, foreign
exchange (FX) markets are more similar to cryptocurrency markets than equity
markets because they also operate 24 hours a day. Of the papers that analyze
95 intraday patterns in FX markets (e.g. Andersen and Bollerslev (1998), Baillie
and Bollerslev (1991), Breedon and Rinaldo (2013), Ito and Hashimoto (2006),
Rinaldo (2009)), two are of particular relevance. Rinaldo (2009) analyzes six
fiat currency pairs and identifies return patterns related to local business hours.
In contrast, Baillie and Bollerslev (1991) report that intraday patterns in FX
100 volatility are similar across currencies and are related to the opening hours of
major financial markets. Our own results are more similar to those of Baillie
and Bollerslev (1991), as we document that intraday patterns are remarkably
similar across trading venues in different continents and time zones. However,
we note that our setting is different because our sample contains data for the
105 same currency pairs from different trading venues. By contrast, all previous
papers on intraday patterns in FX markets analyze only one time series per
currency pair.

Our paper improves our understanding of intraday patterns that are ubiq-
uitous in financial markets. It provides the most comprehensive analysis of in-
110 traday patterns in cryptocurrencies to date. Since there are no trading halts in
cryptocurrency markets, the existence of pronounced intraday patterns in these
markets implies that intraday patterns in returns, trading activity, volatility,
and liquidity are not driven solely by trading frictions such as the overnight and
weekend trading halts. The similarity of intraday patterns across geographic
115 regions and time zones that we document suggests that asset-specific determi-
nants such as the information flow are more important than local factors such
as trader population or trader behavior.

The remainder of the paper is organized as follows. In section 2 we describe
the data used in the empirical analyses, section 3 outlines the methodology. The
120 results are presented in section 4, and section 5 concludes.

2. Data

We source our data from www.cryptotick.com, a commercial vendor of high quality cryptocurrency data. Our data includes hourly observations for opening (O), high (H), low (L), and closing (C) prices as well as the trading volume (V) and the total number transactions (Tx). The sample period extends from July 1 00:00, 2018 to January 1 24:00, 2022 and covers 1,281 trading days (30,744 hours). At the beginning of the sample period, the data set includes 8,598 trading pairs traded on 74 exchanges. Due to the issuance of new cryptocurrencies and the emergence of new trading venues the sample increases over time. At the end of the sample period on January 1, 2022, it includes 25,408 trading pairs traded on 120 exchanges. A "trading pair" is defined at the currency pair - exchange level. If the same currency pair is traded on multiple venues (which is often the case), it will be included several times in our data set.

We apply several filters to our data set. First, for a trading pair to be included in the sample we require that at least 75% (23,058 out of 30,744) hourly observations of O/H/L/C/V/Tx are available. Second, we use the rating information available from [coinmarketcap.com](https://coinmarketcap.com/rankings/exchanges/) and [coingecko.com](https://www.coingecko.com/en/exchanges) which can be retrieved at <https://coinmarketcap.com/rankings/exchanges/> and <https://www.coingecko.com/en/exchanges>, respectively. We discard data from unrated trading venues and data from venues with a poor rating (below 2/10 from both rating providers).

The final data set comprises 1,940 trading pairs traded on 38 exchanges around the globe. Table 1 lists the 38 venues and includes information about their ratings, the number of trading pairs included in our sample, the number of trading pairs that include bitcoin (BTC), ether (ETH), or a fiat currency as one currency, the country where the trading venue is located, and the difference between local time and UTC.

INSERT TABLE 1 ABOUT HERE

Of the 38 trading venues, 10 are located in North and South America, 19

150 in Asia and Oceania, and 9 in Europe. The majority of trading pairs in our sample are traded in Asian trading venues (1,518 out of 1,940), while 254 (168) are traded in venues in the Americas (Europe). We also note that the sample includes 386 different cryptocurrency coins or tokens, including 23 fiat currencies and stablecoins. 46 of them are among the top 100 cryptocurrencies in terms of
155 market capitalization as of February 3, 2022.³

3. Methodology

We use our data on open, high, low and closing prices as well as trading volume and the number of transactions to calculate hourly measures of return, volatility, illiquidity, and trading activity. Specifically, we calculate the following
160 variables.

1. Hourly log returns $R_t = \log(C_t/C_{t-1})$, where C_t denotes the closing price in interval t .

2. The Garman and Klass (1980) volatility estimator

$$\hat{\sigma}_t = \sqrt{0.5 \cdot (\ln(H_t/L_t))^2 - (2 \cdot \ln(2) - 1) \cdot (\ln(C_t/O_t))^2},$$

165 where O_t , H_t , L_t and C_t denote the respective hourly opening, high, low and closing prices in interval t .

3. The Corwin and Schultz (2012) estimator (CS) of the percentage bid-ask spread.⁴ The CS estimator is calculated from the high and low prices of two adjacent hourly intervals $t, t+1$ as

$$170 \quad CS_{t,t+1} = \frac{2(\exp(\alpha)-1)}{1+\exp(\alpha)}, \text{ where} \\ \alpha = (1+\sqrt{2}) \cdot (\sqrt{\beta}-\sqrt{\gamma}), \beta = \left[\ln\left(\frac{H_t}{L_t}\right) \right]^2 + \left[\ln\left(\frac{H_{t+1}}{L_{t+1}}\right) \right]^2, \gamma = \left[\ln\left(\frac{H_{t,t+1}}{L_{t,t+1}}\right) \right]^2.$$

H_t and L_t denote the high and low prices in interval t , while $H_{t,t+1}$ and $L_{t,t+1}$ refer to the high and low price of two adjacent intervals t and

³Some large cryptocurrencies (such as Solana, Polkadot, Terra, Avalanche, etc.) are excluded from our data set due to a lack of time series data. Remember that we require at least 23,058 observations out of a maximum of 30,744 (equivalent to 961 days out of 1,281).

⁴We opt for the Corwin and Schultz (2012) spread estimator because Brauneis et al. (2021) show that it performs best in capturing time series variations in cryptocurrency liquidity.

- 175 $t + 1$. We follow Corwin and Schultz (2012) and set negative values of
the proxy to zero. The CS_t estimator for period t is then calculated as
 $(CS_{t-1,t} + CS_{t,t+1})/2$
4. The total hourly log-trading volume in interval t , $\log(V_t)$
 5. The number of transactions in interval t , Tx_t .
 - 180 6. The average trade size, defined as V_t/Tx_t , in interval t .

The resulting time series are denoted by $X_{t,i}^{(m)}$, where $t = \{1, 2, \dots, 30744\}$ denotes the hourly interval, $i = \{1, 2, \dots, 1940\}$ denotes the trading pair and (m) denotes the respective variable. To facilitate comparison across the time series we detrend them (by removing a linear trend), de-mean them and divide them
185 by their respective standard deviation. We denote the resulting normalized time series as $x_{t,i}^{(m)}$. In the next step we regress, in the tradition of French (1980) and Gibbons and Hess (1981), the normalized time series of hourly observations on a set of dummy variables representing the 24 hours of the trading day, i.e.

$$x_t = \sum_{j=1}^{24} \beta_j \cdot D_j + \epsilon_t. \quad (1)$$

The dummy variable D_j , with $j = \{1, 2, \dots, 24\}$, takes the value 1 if $t \equiv j - 24(\text{mod } -24)$ (in other words, if t is an observation from hour j) and zero otherwise,
190 and ϵ_t is the error term. This regression delivers, for each of the 1,940 trading pairs and six variables, a time series of 24 coefficients which capture the intraday pattern for the trading pair and variable under consideration. These sets of 24 coefficients are the basis for our analysis.

195 4. Results

4.1. Hour of the day patterns

We start by plotting simple averages of the coefficients derived from Eq. 1 for the entire sample. Figure 1 shows the results for each of the six measures.

INSERT FIGURE 1 ABOUT HERE

200 We find that returns tend to be below average over the course of the day from 1:00 to 5:00 UTC and reach their peak between 15:00 and 16:00 UTC. This corresponds to the time window during which the major stock exchanges in the U.S. and Europe are simultaneously open. Returns are also positive in the last three hours of the day.

205 For the remaining five measures, we find very clear and broadly similar intraday patterns. Volatility, illiquidity, trading volume, and the number of transactions are uniformly below average in the first third of the day in UTC (except for the first hour). This corresponds to the period when European and U.S. stock markets are closed. Trading volume and the number of transactions
210 develop very similarly in this interval, an observation which is mirrored in the fact that the average trade size is close to the daily average. Between 08:00 and 12:00 UTC all five variables are close to zero. Thereafter, however, the values for all variables increase steadily, peaking between 16:00 and 17:00 UTC (the only exception being the Corwin/Schultz spread estimator that peaks between 15:00
215 and 16:00 UTC). Obviously, most trading activity in cryptocurrency markets occurs around London tea time (late morning in New York) and is accompanied by high volatility and low market liquidity. The finding that liquidity is low when volume is high confirms the results reported in Brauneis et al. (2022) and Dyhrberg et al. (2018), among others. The observation that trading activity
220 and volatility peak simultaneously is consistent with the Admati and Pfleiderer (1988) model. The remaining hours of the day are characterized by below-average values of volatility and trading activity, as well as increased liquidity.

We next calculate, for each of our six measures of returns, volatility, liquidity and trading activity, simple averages of the time-of-day dummies at the
225 continent level. Figure 2 shows the results.

INSERT FIGURE 2 ABOUT HERE

The intraday patterns bear striking similarities across continents. The lowest returns are always observed in the early morning hours (UTC), the highest returns in the early afternoon and late evening. Volatility, trading volume, the

230 number of transactions and illiquidity have, with few exceptions, similar intra-
day patterns. They have a peak in the first hour of the day (Europe is an
exception here), then are below their respective means until lunchtime, peak in
the afternoon and are again below their means in the evening. Despite these
similarities, there are also patterns which relate to local time, in particular for
235 trading volume and the number of transactions. These variables tend to take
on their lowest values at night time in the respective continent, that is, evening
UTC time for Asia, night time UTC for Europe and early morning UTC for the
Americas. In spite of these regional differences, though, trading activity as well
as volatility and illiquidity peak in the afternoon UTC, at UK tea time.

240 Figure 2 provides initial evidence that the intraday patterns in trading ac-
tivity, volatility and illiquidity that emerge at trading venues around the globe
share many similarities. There is thus pronounced commonality in the intraday
patterns. We note, though, that the intraday patterns are also affected by local
factors such as the time zone the venue is located in. We will provide a more
245 detailed analysis of the intraday patterns in the subsequent sections.

4.2. Correlations

To analyze the extent and the determinants of the commonality in intraday
patterns more formally we need a quantitative measure of the similarities of the
intraday patterns. An intuitive measure is the correlation between the 24 hour-
250 of-day dummies for two trading pairs. All subsequent analyses are therefore
based on the 1940×1940 correlation matrix of dummy coefficient sets. This
matrix holds a total of $1940 \cdot (1940 - 1)/2 = 1,880,830$ unique correlation
coefficients for each of the six measures.

In a first step, we arrange the correlation matrix into a 3-by-3 grid of all
255 trading pairs from American, Asian and European trading venues and then
report the average correlation for each of the nine cells. The results are shown in
Table 2. The figures on the diagonal are the mean correlations for trading pairs
from the same continent while the off-diagonal elements show cross-continent
averages.

INSERT TABLE 2 ABOUT HERE

The correlations are generally lowest for returns and average trade size, both within and across continents. The correlations for volatility, trading volume and the number of transactions are much larger. Unsurprisingly, the within-
 265 continent correlations are always higher than the across-continent correlations. These findings fully confirm our earlier results that there is strong commonality in the intraday patterns of trading activity, volatility and illiquidity, and that local (continent-specific) factors also play a role in shaping the intraday patterns. European trading venues have the highest within-continent correlations for re-
 270 turns, trading volume, and transactions, while trading venues in the Americas have the highest commonality for volatility and liquidity. Across continents, correlations are highest between exchanges in Asia and the Americas (Europe and the Americas) for volatility, liquidity, the number of transactions and trade size (returns and trading volume).

275 In the next step, we examine average correlations at a lower level of aggregation. Specifically, we report exchange-level mean correlations. Figure 3 shows the results for returns, similar figures for the other measures can be found in the online appendix. (Figures 5 to 9).

INSERT FIGURE 3 ABOUT HERE

Green (red) color indicates positive (negative) correlations. More intense colors and larger circles mean higher numerical values of the correlations. The overall color is light green, confirming our earlier evidence of moderately positive correlation between the intraday patterns of returns. However, there are individ-
 285 ual exchanges (such as BTCMARKETS, INDEPENDENTRESERVE, ITBIT, YUNEX, or ZAIF) that have much higher positive correlations with most other trading venues.

Considering the figures for volatility, illiquidity and trading activity in the

appendix, we find much higher average correlations. While this holds for al-
 290 most all exchange pairs for volatility, there are some notable exceptions for the
 intraday patterns in liquidity and trading activity. For these measures there
 are some exchanges that have intraday patterns which are negatively correlated
 with those of other trading venues. An important question that arises in this
 context is whether we can identify determinants of the degree of commonality
 295 in intraday patterns between trading pairs. We address this question in the
 following sections.

4.3. Correlation clustering

So far, we have only considered bivariate correlations between intraday pat-
 terns of two trading pairs. We now go one step further and analyze, separately
 300 for each of our six variables, how similar the correlations are between two trading
 pairs and the 1,938 other trading pairs. For this purpose, we use (agglomera-
 tive) hierarchical cluster analysis. In doing so, we proceed as follows. We start
 from the 1,940 by 1,940 matrix $\mathbf{C}$, which contains the correlations between the
 intraday patterns of our trading pairs, and consider two trading pairs labelled j
 305 and k . Let c_j and c_k be the j^{th} and k^{th} rows of $\mathbf{C}$, respectively. c_j (c_k) contains
 the correlations between the intraday patterns for the j (k) trading pair and all
 1,940 trading pairs in our sample. We then calculate the Euclidean distance⁵
 between c_j and c_k as

$$d_{j,k}^2 = (c_j - c_k) \cdot (c_j - c_k)'.$$

This procedure is repeated for all $1,940 \cdot 1939/2$ distinct trading pairs. The
 310 hierarchical clustering algorithm then selects the two pairs with the lowest dis-
 tance metric into a group A. It then considers the second-lowest distance. If this
 distance is between a trading pair that is not included in group A and one of the
 two pairs in group A, this third pair is added to group A. If the second-lowest

⁵Hierarchical cluster analysis can also be performed with other distance metrics. We choose
 Euclidean distance because of its intuitive appeal.

distance is between two pairs which are both not included in group A, then
 315 these two pairs form a second group B. In each further step, the algorithm can
 either add an element to an existing group, form a new group, or merge two ex-
 isting groups. The latter operation requires a measure of the distance between
 groups.⁶ If the algorithm is not stopped, it continues until all trading pairs
 form a single large group. Two procedures allow the algorithm to be stopped.
 320 Either one defines a threshold for the distance metric and the algorithm stops
 when all remaining distances are above that threshold. Alternatively, one can
 pre-define the number of clusters to be created. We opt for the second approach
 and choose a number of five clusters.⁷

To illustrate the procedure we consider intraday patterns in returns. We
 325 first randomize the correlation matrix $\mathbf{C}$, i.e. we randomly assign the 1,940
 trading pairs to the rows and columns of the matrix.⁸ The randomized cor-
 relation matrix is shown in the left Panel of Figure 4.⁹ The color coding is
 such that positive (negative) correlations are represented by green (red) dots,
 and a more intense color indicates a higher numerical value of the correlation
 330 coefficient. The observation that the matrix is light green confirms our previous
 finding that the intraday patterns in returns are generally positively correlated.
 Next, we deploy the hierarchical cluster analysis. It identifies two large clusters,
 two medium-sized clusters and one very small cluster. The right-hand Panel
 of Figure 4 shows the correlation matrix when we order the trading pairs by
 335 their group membership. The five blocks along the diagonal represent the cor-
 relations within a cluster, while the remaining blocks represent the correlations

⁶There are several alternatives. We opt for the complete linkage procedure (Macnaughton-Smith (1965)), which assigns to two groups a distance equal to the maximum of the distances of the elements within the groups. We have also implemented alternative procedures (e.g. the minimum distance as well as the average distance of the elements within the groups) and found that they yield qualitatively similar results.

⁷We repeated the analysis with values from three to eight and found five to be the most reasonable choice. Allowing more than five groups results in some very small groups, allowing less than five results in one dominating group.

⁸Note that the randomization is not necessary. It is used to better illustrate how hierarchical cluster analysis reveals structure in what is initially a completely unstructured matrix.

⁹Figures similar to Figure 4 for the other five variables are shown in the online appendix, Figures 10 to 14.

between clusters. Visual inspection of the right-hand Panel of Figure 4 reveals the structure identified by cluster analysis. For example, the trading pairs in cluster 1 have a high correlation within the cluster, but a low or even negative correlation with the trading pairs in clusters 4 and 5.

INSERT FIGURE 4 ABOUT HERE

It is interesting to see *which* trading pairs are grouped together. We take into account the location of the trading venue (America, Asia, Europe) as well as whether a trading pair contains bitcoin, ether, a fiat currency, or a stablecoin as one currency. For each cluster, the first five rows of Table 3 show the number of trading pairs traded on a trading venue in the Americas, Asia, and Europe, as well as the number of pairs that contain bitcoin, ether, a fiat currency or a stablecoin as one currency. In addition to the actual numbers, the table also indicates (in parentheses) the numbers that would be expected if the trading pairs were randomly assigned to clusters of the respective size.

INSERT TABLE 3 ABOUT HERE

The table shows clear patterns. Cluster one (with 713 trading pairs) consists mainly of pairs containing a fiat currency or a stablecoin. 539 pairs exhibit this characteristic. If the pairs were randomly assigned to a cluster of size 713, we would expect only 290 pairs with this property. Cluster 1 also contains a disproportionately large number of pairs from exchanges in the Americas and Europe, while pairs from Asian exchanges are underrepresented. In addition, a disproportionately small number of pairs in cluster 1 contain bitcoin or ether while the opposite is true for trading pairs in cluster 2. Thus, we conclude that the similarity of intraday return patterns is determined by the location of the trading venue and the presence of bitcoin, ether or a fiat currency or stablecoin in the trading pair.

Table 3 also shows results of the hierarchical cluster analysis for the other variables (volatility, spreads, trading volume, number of trades and average trade size). They are consistent with the results for returns in that they also

suggest that the location of the exchange is a determinant of the intraday patterns. However, unlike the results for returns, the numbers of pairs containing bitcoin, ether, or a fiat currency or a stablecoin do not deviate significantly from their expected values.

370 4.4. Drivers of intraday patterns

The hierarchical cluster analysis presented in the previous section allows us to draw conclusions about the determinants of intraday patterns by considering the properties of trading pairs in the resulting clusters. A regression analysis can deliver additional insights. We use the 1940 · 1939/2 correlations between
 375 the intraday patterns for the trading pairs as the dependent variable. The independent variables are dummy variables that identify (1) trading pairs traded on venues on the same continent (variable *Cont*), (2) trading pairs that trade on the same trading venue (*Venue*), (3) identical currency pairs traded on different venues (*Pair*), (4) trading pairs that share one currency (*Share*), (5) pairs that
 380 both contain bitcoin (*BBTC*), (6) pairs that both contain ether (*BETH*), (7) pairs that both contain a fiat currency or a stablecoin (*BFiat*). It is well known that some cryptocurrency exchanges report inflated trading volume figures (e.g. Cong et al. (2022)). The fake volume that is reported may, in turn, affect the intraday patterns. We therefore include two additional dummy variables to
 385 capture the reliability of exchange reporting. For this purpose we rely on the exchange rating provided by coinmarketcap.com and categorize trading venues with a score above 6 as reliable.¹⁰ We then define the dummy variables *BTrust* and *1Trust* which are set to one when both venues and one venue, respectively, in a pair are traded on a reliable venue. Finally, we add as an independent vari-
 390 able the absolute time difference, measured in hours, between the two trading venues (*Time*), resulting in the following specification (the subscripts have been omitted for ease of notation).

¹⁰ An alternative to this ranking is the "trust score" provided by coingecko.com. All venues which have a score above 6 in the coinmarketcap.com rating also have a coingecko trust score above 6. However, coingecko assigns scores above 6 to a much larger number of venues.

$$\hat{\rho} = c + \beta_1 Cont + \beta_2 Venue + \beta_3 Pair + \beta_4 Share + \beta_5 BBTC + \\ + \beta_6 BETH + \beta_7 BFiat + \beta_8 Time + \beta_9 BTrust + \beta_{10} ITrust + \epsilon \quad (2)$$

We estimate one such regression for each of our six dependent variables. The results are shown in Table 4. The constants (representing estimates of the correlation of intraday patterns for a hypothetical trading pair for which all independent variables are zero) are positive and reflect our earlier results - the correlation of intraday patterns is highest for volatility, intermediate for spreads, trading volume, and the number of transactions, and lowest for returns and average trade size. The coefficient estimates of the dummy variables have an intuitive interpretation. They provide an estimate of the change in the correlation of the intraday patterns when the value of the respective dummy variable is increased from 0 to 1. Due to the large number of observations, all coefficient estimates are statistically significant. When interpreting our results we therefore emphasize the economic significance of the estimates.

INSERT TABLE 4 ABOUT HERE

The coefficient of the dummy variable *Cont* is positive for five out of six measures (the exception being the number of transactions), suggesting that the intraday patterns of two pairs are generally more highly correlated when the pairs are traded on venues on the same continent. The effects are particularly strong for trading volume and trade size, and are negligible for returns and the number of transactions. The coefficient for *Venue* is always positive, implying that pairs traded on the same trading venues have more similar intraday patterns. The effects are strong for all dependent variables, with coefficient estimates ranging from 0.05 (returns) to 0.26 (volume). Similarly, the intraday patterns of the same currency pair traded on different venues are more highly correlated than those of different currency pairs. The effects are again very

strong, with coefficient estimates between 0.06 (trade size) and 0.58 (returns). The same is true for pairs that share a common currency. However, the coefficients are markedly lower than the coefficients for identical currency pairs, with values ranging between 0.007 (returns) and 0.06 (number of transactions). Interestingly, the coefficients for *BBTC* and *BETH* are mostly negative while those for *BFiat* are mostly positive.¹¹ Thus, if two pairs both include bitcoin or ether as one currency, then the intraday patterns tend to be less similar. On the other hand, if both pairs contain a fiat currency or a stablecoin, the intraday patterns tend to be more similar. The coefficients for the dummy variables that capture trading venue reliability are positive for five of our six measures (the exception being returns) and are of an economically significant magnitude for the Corwin and Schultz (2012) spread estimator and the measures of trading activity (volume, number of trades and trade size). The coefficient estimate in the return regression is numerically small and negative, implying that intraday patterns in returns generated on reliable exchanges are, if anything, less similar than those from unreliable exchanges. A potential explanation may be that venues reporting fake transactions (or wash trades) report prices "borrowed" from other venues, thereby artificially creating intraday patterns of returns that resemble the patterns generated on other venues. Finally, the coefficients for the time difference variable are predominantly negative, suggesting that the intraday patterns of two trading pairs are more highly correlated when the time difference between the locations of the trading venues is lower. The only exception is the intraday pattern in volatility. Here, the coefficient of the time difference variable is positive, but, because of its small size, economically negligible.

It is worth noting that the intraday patterns in volatility are indeed strongly positively correlated. Given that all coefficients except *BBTC* are positive in the volatility regression, the lowest predicted value (0.453) obtains when *Share* and

¹¹Note that, when two trading pairs contain bitcoin or ether, the variable *Share* is also set to 1. Thus, the total effect when bitcoin or ether are contained in two pairs is obtained by adding the coefficients for the *Share* variable and *BBTC* or *BETH*. The same is true for the variable *BFiat* when the two trading pairs under consideration contain the same fiat currency.

445 BBTC are set to one¹² and all other independent variables are set to zero. This estimate implies that the (predicted) correlation between the intraday patterns in volatility never drops below 45.3%, even when the two trading pairs have nothing in common and are traded on different venues located on different continents. This finding suggests a high degree of commonality in cryptocurrency
450 volatility.

Overall, the regression results confirm our earlier findings that intraday patterns are generally positively correlated, that the correlation is highest for volatility, followed by our measures of the spread and trading activity, and that it is lowest for returns and average trade size. We also find that correlations
455 increase when the venues where trading pairs are traded are located on the same continent or in closer time zones, and when trading pairs share a common currency, or are traded on the same venue.

5. Conclusion

In this paper we analyze intraday patterns in trading activity, returns,
460 volatility and liquidity for a large sample of cryptocurrencies traded on 38 trading venues all over the world. What is special about this setting is that (1) there are no institutional frictions such as overnight or weekend trading halts in cryptocurrency markets and (2) that the same currency pairs are traded on multiple venues located in different continents and different time zones. Any
465 intraday patterns that we find can thus not be caused by institutional frictions. Further, to the extent that intraday patterns are similar across continents and time zones, we can rule out local habits (such as lunchtime conventions) as a cause of the joint pattern.

We find pronounced intraday patterns in trading activity, volatility and illiquidity.
470 These patterns are remarkably similar to each other. Trading activity, volatility and illiquidity are lowest in the early morning hours (UTC) and are

¹²Note that Share must have the value 1 when BBTC is set to 1.

highest in the afternoon, around UK tea time. In addition, the intraday patterns are also similar across geographic regions (America, Asia, Europe). The similarity is most pronounced for volatility.

475 We then estimate regressions where the dependent variable is the correlation in intraday patterns between two trading pairs (i.e., currency A traded against currency B on venue C versus currency D traded against currency E on venue F where A and D (B and E, C and F) may or may not be equal). We find that the intraday patterns are more highly correlated when the pairs are more
480 similar (e.g. share a common currency), or when the pairs are traded on the same venue or on venues on the same continent and/or in the same time zone.

Our results suggest that pronounced intraday patterns emerge endogenously, i.e. absent institutional frictions. They further imply that these patterns are not fully explained by local factors but that there rather exists a pronounced
485 commonality in intraday patterns.

References

- Admati, A., Pfleiderer, P., 1988. A Theory of Intraday Patterns: Volume and Price Variability. *Review of Financial Studies* 1, 3–40. doi:<https://doi.org/10.1093/rfs/1.1.3>.
- 490 Aharon, D.Y., Qadan, M., 2019. Bitcoin and the day-of-the-week effect. *Finance Research Letters* 31, 415–424. doi:<https://doi.org/10.1016/j.frl.2018.12.004>.
- Andersen, T.G., Bollerslev, T., 1998. Deutsche mark–dollar volatility: Intra-day activity patterns, macroeconomic announcements, and longer run dependencies. *Journal of Finance* 53, 219–265. doi:<https://doi.org/10.1111/0022-1082.85732>.
495
- Aslanidis, N., Bariviera, A.F., Perez-Laborda, A., 2021. Are cryptocurrencies becoming more interconnected? *Economics Letters* 199, 109725. doi:<https://doi.org/10.1016/j.econlet.2021.109725>.
- 500 Baillie, R.T., Bollerslev, T., 1991. Intra-day and inter-market volatility in foreign exchange rates. *Review of Economic Studies* 58, 565–585. doi:<https://doi.org/10.2307/2298012>.
- Baur, D.G., Cahill, D., Godfrey, K., (Frank) Liu, Z., 2019. Bitcoin time-of-day, day-of-week and month-of-year effects in returns and trading volume. *Finance Research Letters* 31, 78–92. doi:<https://doi.org/10.1016/j.frl.2019.04.023>.
505
- Bouri, E., Vo, X.V., Saeed, T., 2021. Return equicorrelation in the cryptocurrency market: Analysis and determinants. *Finance Research Letters* 38, 101497. doi:<https://doi.org/10.1016/j.frl.2020.101497>.
- 510 Brauneis, A., Mestel, R., Riordan, R., Theissen, E., 2021. How to Measure the Liquidity of Cryptocurrencies. *Journal of Banking and Finance* 124, 106041. doi:<https://doi.org/10.1016/j.jbankfin.2020.106041>.

- Brauneis, A., Mestel, R., Riordan, R., Theissen, E., 2022. Bitcoin unchained: Determinants of cryptocurrency exchange liquidity. *Journal of Empirical Finance* 69, 106–122. doi:<https://doi.org/10.1016/j.jempfin.2022.08.004>.
515
- Breedon, F., Rinaldo, A., 2013. Intraday patterns in fx returns and order flow. *Journal of Money, Credit and Banking* 45, 953–965. doi:<https://doi.org/10.1111/jmcb.12032>.
- Caporale, G.M., Plastun, A., 2019. The day of the week effect in the cryptocurrency market. *Finance Research Letters* 31, 258–269. doi:<https://doi.org/10.1016/j.frl.2018.11.012>.
520
- Cong, L., Li, X., Tang, K., Yang, Y., 2022. Crypto Wash Trading. NBER Working Paper 30783 30783. doi:10.3386/w30783.
- Corwin, S.A., Schultz, P., 2012. A Simple Way to Estimate Bid-Ask Spreads from Daily High and Low Prices. *Journal of Finance* 67, 719–760. doi:10.1111/j.1540-6261.2012.01729.x.
525
- Dorflleitner, G., Lung, C., 2018. Cryptocurrencies from the perspective of euro investors: a re-examination of diversification benefits and a new day-of-the-week effect. *Journal of Asset Management* 19, 472–494. doi:<https://doi.org/10.1057/s41260-018-0093-8>.
530
- Dyhrberg, A., Foley, S., Svec, J., 2018. How Investible Is Bitcoin? Analyzing the Liquidity and Transaction Costs of Bitcoin Markets. *Economics Letters* 171, 140–143. doi:<https://doi.org/10.1016/j.econlet.2018.07.032>.
- Eross, A., McGroarty, F., Urquhart, A., Wolfe, S., 2019. The intraday dynamics of bitcoin. *Research in International Business and Finance* 49, 71–81. doi:<https://doi.org/10.1016/j.ribaf.2019.01.008>.
535
- French, K.R., 1980. Stock returns and the weekend effect. *Journal of Financial Economics* 8, 55–69. doi:[https://doi.org/10.1016/0304-405X\(80\)90021-5](https://doi.org/10.1016/0304-405X(80)90021-5).
540

- Garman, M., Klass, M., 1980. On the Estimation of Security Price Volatilities from Historical Data. *The Journal of Business* 53, 67–68. doi:<https://doi.org/10.1086/296072>.
- Gibbons, M.R., Hess, P., 1981. Day of the week effects and asset returns. *Journal of Business* 54, 579–596. doi:<https://dx.doi.org/10.1086/296147>.
545
- Hasan, M., Naeem, M.A., Arif, M., Shahzad, S.J.H., Vo, X.V., 2022. Liquidity connectedness in cryptocurrency market. *Financial Innovation* 8, 3. doi:<https://doi.org/10.1186/s40854-021-00308-3>.
- Heston, S.L., Korajczyk, R.A., Sadka, R., 2010. Intraday patterns in the cross-section of stock returns. *Journal of Finance* 65, 1369–1407. doi:<https://doi.org/10.1111/j.1540-6261.2010.01573.x>.
550
- Hu, A., Parlour, C.A., Rajan, U., 2019. Cryptocurrencies: Stylized facts on a new investible instrument. *Financial Management* 48, 1049–1068. doi:<https://doi.org/10.1111/fima.12300>.
- Ito, T., Hashimoto, Y., 2006. Intraday seasonality in activities of the foreign exchange markets: Evidence from the electronic broking system. *Journal of the Japanese and International Economies* 20, 637–664. doi:<https://doi.org/10.1016/j.jjie.2006.06.005>.
555
- Kaiser, L., 2019. Seasonality in cryptocurrencies. *Finance Research Letters* 31, 232–238. doi:<https://doi.org/10.1016/j.frl.2018.11.007>.
560
- Kinateder, H., Papavassiliou, V.G., 2021. Calendar effects in bitcoin returns and volatility. *Finance Research Letters* 38, 101420. doi:<https://doi.org/10.1016/j.frl.2019.101420>.
- Long, H., Zaremba, A., Demir, E., Szczygielski, J.J., Vasenin, M., 2020. Seasonality in the cross-section of cryptocurrency returns. *Finance Research Letters* 35, 101566. doi:<https://doi.org/10.1016/j.frl.2020.101566>.
565

- Ma, D., Tanizaki, H., 2019. The day-of-the-week effect on bitcoin return and volatility. *Research in International Business and Finance* 49, 127–136. doi:<https://doi.org/10.1016/j.ribaf.2019.02.003>.
- 570 Macnaughton-Smith, P., 1965. Some Statistical and Other Numerical Techniques for Classifying Individuals. Research Unit Report 6. London: Her Majesty's Stationary Office .
- McInish, T.H., Wood, R.A., 1992. An analysis of intraday patterns in bid/ask spreads for nyse stocks. *Journal of Finance* 47, 753–764. doi:<https://doi.org/10.1111/j.1540-6261.1992.tb04408.x>.
- 575
- Petukhina, A.A., Reule, R.C., Haerdle, K., 2021. Rise of the machines? intraday high-frequency trading patterns of cryptocurrencies. *European Journal of Finance* 27, 8–30. doi:<https://doi.org/10.1080/1351847X.2020.1789684>.
- Qadan, M., Aharon, D.Y., Eichel, R., 2021. Seasonal and calendar effects and the price efficiency of cryptocurrencies. *Finance Research Letters* 46, 102354. doi:<https://doi.org/10.1016/j.frl.2021.102354>.
- 580
- Ranaldo, A., 2009. Segmentation and time-of-day patterns in foreign exchange markets. *Journal of Banking and Finance* 33, 2199–2206. doi:<https://doi.org/10.1016/j.jbankfin.2009.05.019>.
- Shams, A., 2020. The structure of cryptocurrency returns. Fisher College of Business Working Paper No. 2020-03-011. URL: <https://ssrn.com/abstract=3604322>.
- 585
- Su, F., Wang, X., Yuan, Y., 2022. The intraday dynamics and intraday price discovery of bitcoin. *Research in International Business and Finance* 60, 101625. doi:<https://doi.org/10.1016/j.ribaf.2022.101625>.
- 590
- Tripathi, A., Dixit, A., Vipul, 2021. Return equicorrelation in the cryptocurrency market: Analysis and determinants. *Applied Economics* 38, 101497. doi:<https://doi.org/10.1016/j.frl.2020.101497>.

Wood, R.A., McInish, T.H., Ord, J.K., 1985. An investigation of transactions
595 data for nyse stocks. *Journal of Finance* 40, 723–739. doi:[https://doi.org/
10.1111/j.1540-6261.1985.tb04996.x](https://doi.org/10.1111/j.1540-6261.1985.tb04996.x).

TABLES

	Exchange	CMC	CG	Pairs	BTC	ETH	Fiat	Country	UTC
1	BIBOX	3.8	7	49	17	18	17	Hong Kong	8
2	BINANCE	9.9	10	284	120	75	60	China	8
3	BITBANK	5.1	9	5	2	1	4	Japan	9
4	BITFINEX	7.3	10	52	22	8	33	Hong Kong	8
5	BITFOREX	4.2	7	42	10	11	24	Hong Kong	8
6	BITHUMB	6.5	8	49	1	1	49	South Korea	9
7	BITSO	4.7	10	10	3	2	8	Mexico	-7
8	BITSTAMP	7.0	9	15	6	3	11	Luxembourg	1
9	BITTREX	6.0	10	93	65	11	23	US Seattle	-8
10	BTCBOX	3.5	6	1	1	0	1	Japan	9
11	BTCMARKETS	N/A	6	4	1	1	4	Australia	11
12	BW	2.4	5	8	1	1	8	Australia	11
13	CEXIO	4.5	7	13	6	3	11	UK	0
14	COINBASE	8.4	10	40	12	5	32	US San Fran	-7
15	COINFLOOR	2.9	N/A	1	1	0	1	UK	0
16	COINMATE	3.8	N/A	2	2	0	2	UK	0
17	COINONE	6.4	7	8	1	1	8	South Korea	9
18	CREX24	2.6	4	3	3	1	0	Cyprus	2
19	EXMO	4.4	9	96	29	12	62	UK	0
20	GATEIO	7.5	10	82	8	8	69	China	8
21	GEMINI	7.1	10	6	2	2	5	US New York	-5
22	HITBTC	4.2	6	140	64	36	47	Hong Kong	8
23	HUOBIPRO	7.4	10	400	184	137	74	Hong Kong	8
24	INDEPENDENTRESERVE	4.1	7	2	1	1	2	Australia	11
25	ITBIT	0.1	4	2	1	1	2	US	-7
26	KRAKEN	7.9	10	54	20	6	35	US San Fran	-7
27	KUCCOIN	7.7	10	142	92	32	22	Singapore	8
28	LUNO	4.5	7	4	4	0	4	Singapore	8
29	MERCA DOBITCOIN	4.3	7	2	1	0	2	Brazil	-3
30	OKEX	6.6	10	192	70	34	89	Hong Kong	8
31	POLONIEX	6.3	8	43	24	5	19	US San Fran	-7
32	SOUTHXCHANGE	2.3	4	2	2	0	0	Argentina	-3
33	THEROCKTRADING	3.8	6	5	2	1	4	Italy	1
34	TIDEX	4.0	6	2	2	1	0	US San Fran	-7
35	UPBIT	6.1	8	51	4	1	49	South Korea	9
36	YOBIT	2.4	4	29	12	4	18	Russia	3
37	YUNEX	1.7	3	2	1	0	2	China	8
38	ZAIF	5.8	7	5	1	1	5	Japan	9
					1940	798	424	80	

Table 1: **Properties of exchanges in the sample.** This table presents details on the exchanges used in the empirical analyses. The table presents each exchange's coinmarket-cap.com's spot exchange rating (CMC, max 10) and coingecko.com's exchange trust score (CG, max 10) as of April 13, 2022. Furthermore, it shows the number of total trading pairs in the data set coming from each exchange (Pairs), the number of pairs including bitcoin (BTC) and fiat or tokenized fiat (Fiat) as one of the currencies in each pair. The last two columns report the country of the respective exchange (according to cryptocompare.com) and the time zone relative to coordinated universal time (UTC).

	Returns			Garman/Klass		
	America	Asia	Europe	America	Asia	Europe
America	0.2056	0.1385	0.2229	0.6227	0.5404	0.4979
Asia		0.1393	0.1413		0.5489	0.4167
Europe			0.2521			0.6178

	Corwin/Schultz			Volume		
	America	Asia	Europe	America	Asia	Europe
America	0.4550	0.3723	0.3230	0.5707	0.3535	0.3716
Asia		0.3949	0.2517		0.5321	0.3531
Europe			0.3865			0.6593

	Transactions			Trade Size		
	America	Asia	Europe	America	Asia	Europe
America	0.5139	0.3898	0.3822	0.1336	0.1200	-0.0130
Asia		0.4277	0.3513		0.2241	0.0385
Europe			0.7333			0.0349

Table 2: **Average correlations of dummy variable sets.** This table reports average correlations of dummy coefficient sets for tradings pairs in American, Asian and European exchanges and for each of the six figures - returns, volatility, liquidity, volume, transactions and trade size respectively. The value 0.2056 in column 1 / line 1 denotes the average correlation of dummy coefficient sets for returns for all pairs traded in American venues, while 0.2229 in column 3/line 1 denotes the average correlation of dummy coefficient sets for returns of American vs. European trading pairs.

Cluster	Size	America	Asia	Europe	BTC	ETH	Fiat
Returns 1	713	122 (93)	489 (558)	102 (62)	193 (282)	92 (156)	539 (290)
2	620	41 (81)	563 (485)	16 (54)	311 (245)	237 (136)	59 (252)
3	375	39 (49)	314 (293)	22 (32)	180 (148)	57 (82)	136 (153)
4	210	49 (27)	139 (164)	22 (18)	104 (83)	34 (46)	47 (86)
5	22	3 (3)	13 (17)	6 (2)	10 (9)	4 (5)	9 (9)
Garman/Klass 1	1420	184 (186)	1145 (1111)	91 (123)	524 (562)	316 (310)	629 (578)
2	262	35 (34)	166 (205)	61 (23)	159 (104)	44 (57)	75 (107)
3	195	15 (26)	174 (153)	6 (17)	88 (77)	49 (43)	65 (79)
4	33	2 (4)	21 (26)	10 (3)	9 (13)	9 (7)	14 (13)
5	30	18 (4)	12 (23)	0 (3)	18 (12)	6 (7)	7 (12)
Corwin/Schultz 1	1316	154 (172)	1129 (1030)	33 (114)	471 (521)	293 (288)	590 (536)
2	412	69 (54)	243 (322)	100 (36)	234 (163)	74 (90)	133 (168)
3	88	15 (12)	68 (69)	5 (8)	42 (35)	30 (19)	16 (36)
4	79	8 (10)	57 (62)	14 (7)	30 (31)	21 (17)	29 (32)
5	45	8 (6)	21 (35)	16 (4)	21 (18)	6 (10)	22 (18)
Volume 1	1297	79 (170)	1124 (1015)	94 (112)	502 (513)	305 (283)	503 (528)
2	485	149 (64)	273 (380)	63 (42)	263 (192)	101 (106)	173 (198)
3	104	1 (14)	97 (81)	6 (9)	12 (41)	7 (23)	92 (42)
4	50	25 (7)	21 (39)	4 (4)	21 (20)	9 (11)	21 (20)
5	4	0 (1)	3 (3)	1 (0)	0 (2)	2 (1)	1 (2)
Transactions 1	1483	197 (194)	1130 (1160)	156 (128)	632 (587)	300 (324)	601 (604)
2	328	22 (43)	297 (257)	9 (28)	107 (130)	92 (72)	142 (134)
3	85	33 (11)	52 (67)	0 (7)	42 (34)	19 (19)	28 (35)
4	26	0 (3)	26 (20)	0 (2)	11 (10)	9 (6)	7 (11)
5	18	2 (2)	13 (14)	3 (2)	6 (7)	4 (4)	12 (7)
Trade Size 1	812	60 (106)	737 (635)	15 (70)	298 (321)	183 (177)	348 (331)
2	523	69 (68)	398 (409)	56 (45)	237 (207)	118 (114)	192 (213)
3	278	97 (36)	168 (218)	13 (24)	157 (110)	57 (61)	82 (113)
4	244	18 (32)	151 (191)	75 (21)	71 (97)	52 (53)	132 (99)
5	83	10 (11)	64 (65)	9 (7)	35 (33)	14 (18)	36 (34)
Total values	1940	254	1518	168	768	424	790

Table 3: **Results from hierarchical clustering.** This table reports properties of the clusters derived from the correlation matrices of the dummy variable sets for each of the six measures returns, Garman/Klass, Corwin/Schultz, volume, number of transactions, and the average trade size respectively. Column 2 (size) refers to the number of trading pairs in each cluster, columns 3 to 8 report the number of trading pairs in the respective cluster with respect to the location of the trading venue (America, Asia, Europe) and whether the trading pairs features bitcoin (BTC), ether (ETH) and a fiat currency / stablecoin (Fiat) as the base / quote currency, respectively. Numbers in parentheses report expected values if trading pairs were assigned randomly to the clusters.

Variable	Returns	Garman/Klass	Corwin/Schultz	Volume	Transactions	Trade Size
Constant	0.0908 (0.0010)	0.4832 (0.0011)	0.2280 (0.0011)	0.3755 (0.0013)	0.3941 (0.0013)	0.0527 (0.0015)
Cont	0.0098 (0.0005)	0.0324 (0.0005)	0.0271 (0.0006)	0.1198 (0.0007)	-0.0083 (0.0006)	0.0970 (0.0007)
Venue	0.0527 (0.0007)	0.1804 (0.0008)	0.1807 (0.0008)	0.2582 (0.0009)	0.1959 (0.0009)	0.2220 (0.0010)
Pair	0.5843 (0.0038)	0.2801 (0.0041)	0.3570 (0.0044)	0.2470 (0.0051)	0.3152 (0.0048)	0.0630 (0.0056)
Share	0.0071 (0.0007)	0.0316 (0.0007)	0.0418 (0.0008)	0.0590 (0.0009)	0.0598 (0.0008)	0.0128 (0.0010)
BBTC	0.0120 (0.0008)	-0.0614 (0.0008)	-0.0552 (0.0009)	-0.0431 (0.0010)	-0.0556 (0.0009)	-0.0019 (0.0011)
BETH	-0.0194 (0.0010)	0.0045 (0.0011)	-0.0383 (0.0011)	-0.0538 (0.0013)	-0.0898 (0.0012)	-0.0220 (0.0015)
BFiat	0.3027 (0.0006)	0.0561 (0.0006)	0.0997 (0.0007)	-0.0272 (0.0008)	0.0491 (0.0007)	0.0278 (0.0009)
Time	-0.0002 (0.0000)	0.0003 (0.0000)	-0.0007 (0.0000)	-0.0037 (0.0000)	-0.0043 (0.0000)	-0.0018 (0.0000)
BTrust	-0.0133 (0.0009)	0.0042 (0.0010)	0.1474 (0.0010)	0.0397 (0.0012)	0.0472 (0.0011)	0.0454 (0.0013)
1Trust	-0.0179 (0.0010)	0.0020 (0.0010)	0.0685 (0.0011)	0.0395 (0.0013)	0.0240 (0.0012)	0.0521 (0.0014)
Adj. R2	0.1690	0.0679	0.0873	0.1140	0.0587	0.0578
Observations	1,880,830					

Table 4: **Results correlation regression.** This table reports results from the regression defined in Eq. 2. Dependent variables are correlation coefficients of the dummy variable sets (columns 2 to 7). Independent variables are dummy variables indicating if two pairs: (1) are traded in venues of the same continent (Cont), (2) are traded in the same exchange (Venue), (3) are exactly the same trading pair (Pair), (4) share one common currency (Share), (5) both feature bitcoin (BBTC), (6) both feature ether (BETH), (7) both feature a stablecoin or fiat (BFiat). (8) Time refers to the time gap in hours of the trading venues, (9) BTrust denotes if both pairs are traded in trusted exchanges, and (10) 1Trust indicates that exactly one pair is traded in a trusted exchange. Standard errors in parentheses, all coefficients are significant at the 1% level.

FIGURES

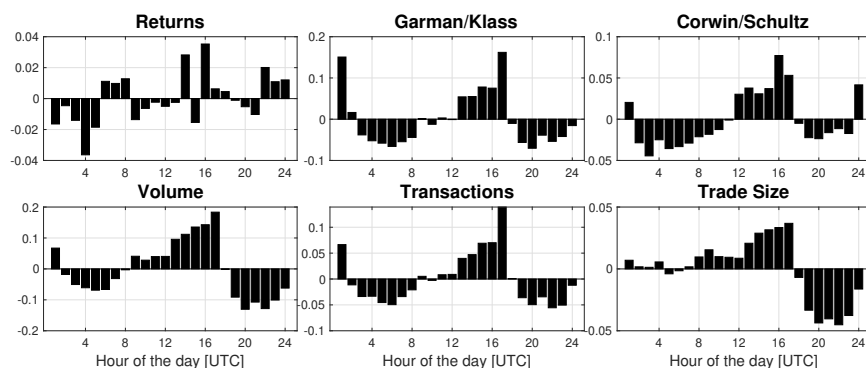


Figure 1: **Time of the day patterns.** This figure reports simple averages of the coefficients of the dummy variable regressions defined in Eq. (1) for the six measures returns, volatility (Garman/Klass), liquidity (Corwin/Schultz), log-trading volume, the number of transactions and the average trade size. Hours refer to Coordinated Universal Time (UTC). E.g., hour 4 covers the period 03:00 to 04:00 UTC.

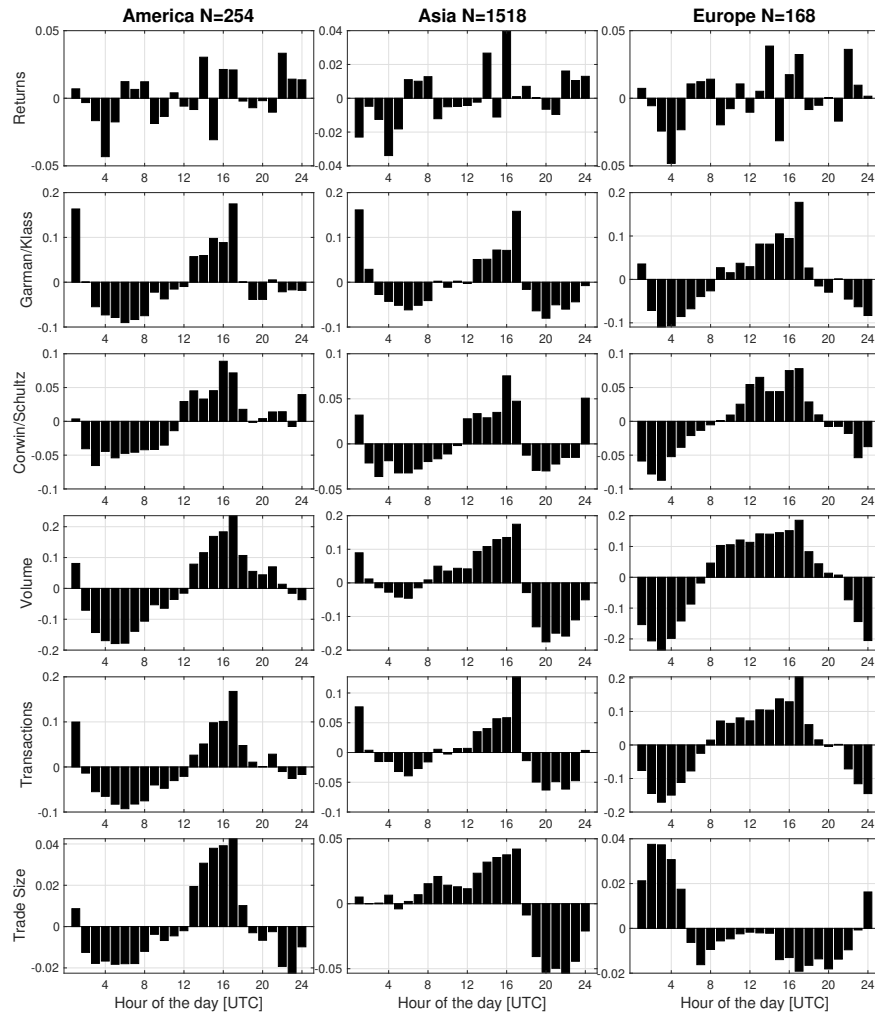


Figure 2: **Time of the day patterns per continent.** This figure reports simple averages of the coefficients of the dummy variable regressions defined in equation (1) for the six measures: returns, volatility (Garman/Klass), liquidity (Corwin/Schultz), log-trading volume, the number of transactions and the average trade size. Results are grouped by location, i.e. the continent in which the headquarters of the respective stock exchange is located. Hours refer to Coordinated Universal Time (UTC). E.g., hour 4 covers the period 03:00 to 04:00 UTC.

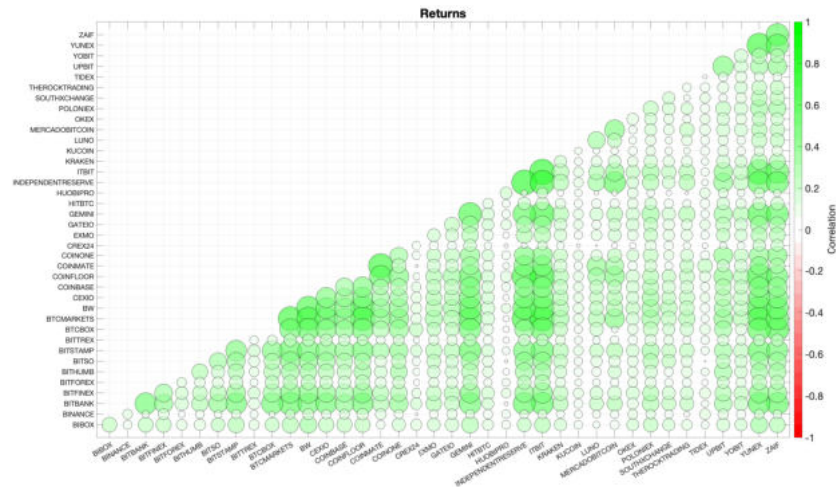


Figure 3: **Average returns correlations across exchanges.** This figure shows average correlations of dummy variable sets for each exchange (diagonal bubbles in the plot) and average correlations of dummy variable sets across individual exchanges (off-diagonal bubbles in the plot).

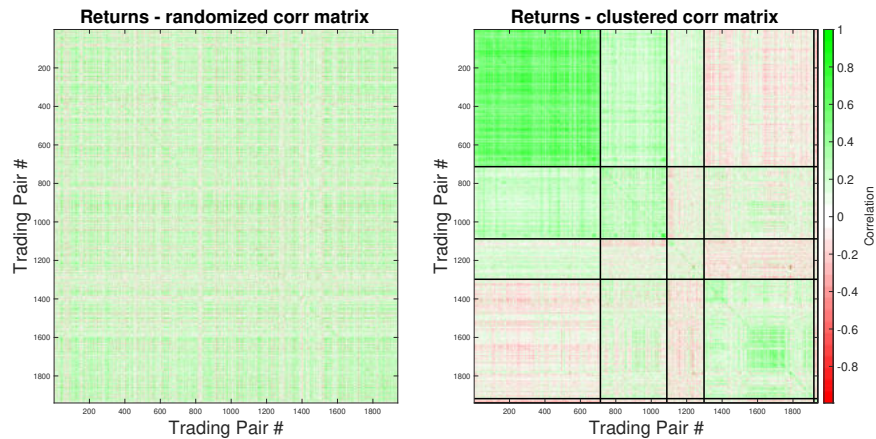


Figure 4: **Hierarchical clustering correlation matrix - Returns.** This Figure plots the randomized correlation matrix for returns (left subplot) and the corresponding clustered correlation matrix (right subplot) with five distinct clusters.

Figures for online Appendix

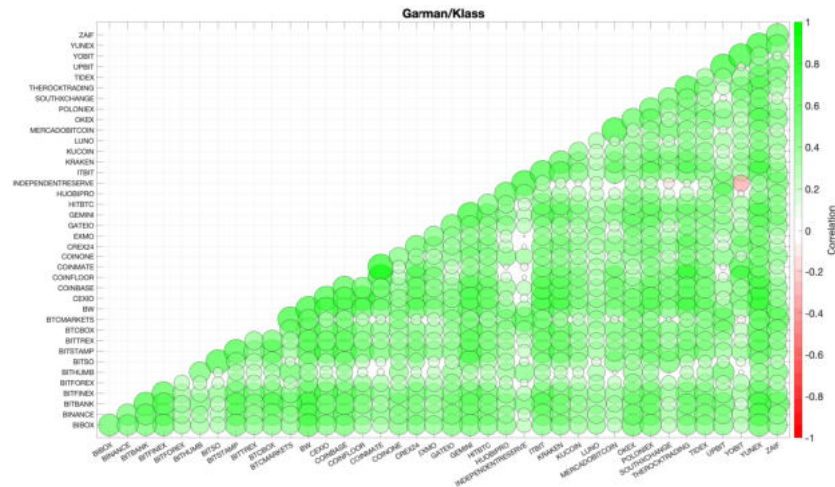


Figure 5: **Average volatility correlations across exchanges.** This figure shows average correlations of dummy variable sets for each exchange (diagonal bubbles in the plot) and average correlations of dummy variable sets across individual exchanges (off-diagonal bubbles in the plot).

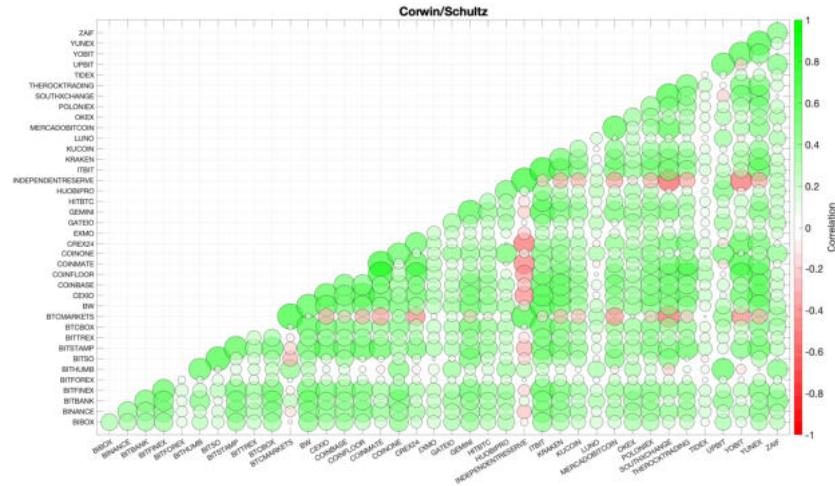


Figure 6: **Average liquidity correlations across exchanges.** This figure shows average correlations of dummy variable sets for each exchange (diagonal bubbles in the plot) and average correlations of dummy variable sets across individual exchanges (off-diagonal bubbles in the plot).



Figure 7: **Average volume correlations across exchanges.** This figure shows average correlations of dummy variable sets for each exchange (diagonal bubbles in the plot) and average correlations of dummy variable sets across individual exchanges (off-diagonal bubbles in the plot).

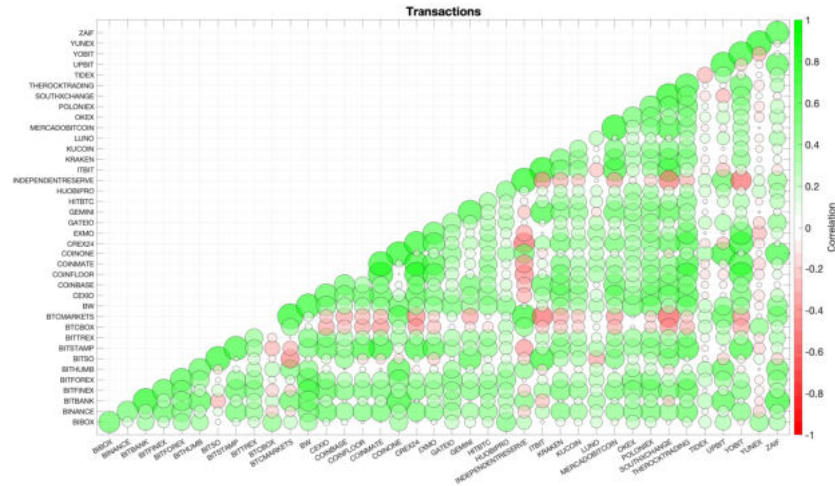


Figure 8: **Average transactions correlations across exchanges.** This figure shows average correlations of dummy variable sets for each exchange (diagonal bubbles in the plot) and average correlations of dummy variable sets across individual exchanges (off-diagonal bubbles in the plot).

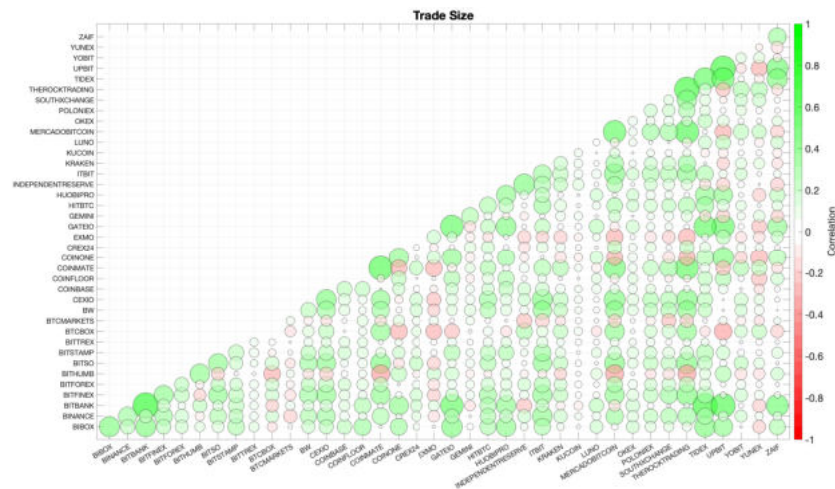


Figure 9: **Average trade size correlations across exchanges.** This figure shows average correlations of dummy variable sets for each exchange (diagonal bubbles in the plot) and average correlations of dummy variable sets across individual exchanges (off-diagonal bubbles in the plot).

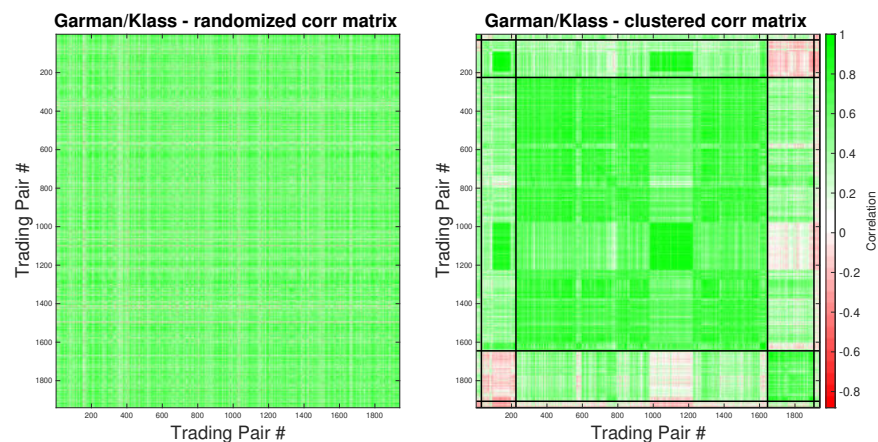


Figure 10: **Hierarchical clustering correlation matrix - Garman/Klass.** This Figure plots the randomized correlation matrix for the Garman/Klass volatility estimator (left subplot) and the corresponding clustered correlation matrix (right subplot) with five distinct clusters.

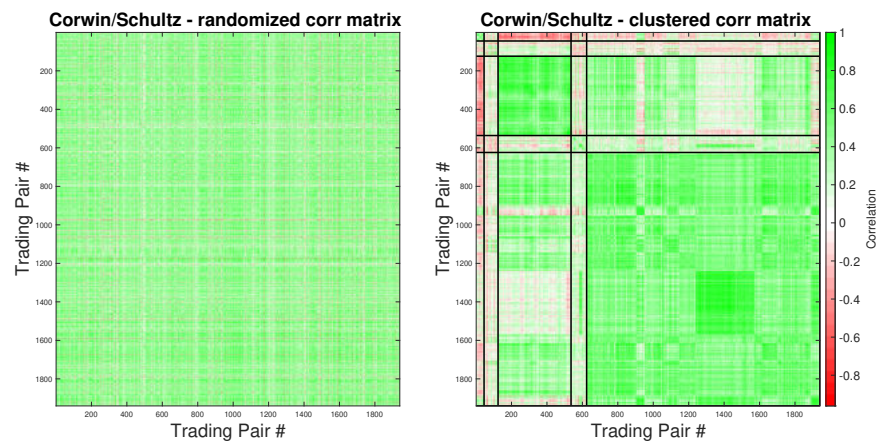


Figure 11: **Hierarchical clustering correlation matrix - Corwin/Schultz.** This Figure plots the randomized correlation matrix for the Corwin/Schultz spread estimator (left subplot) and the corresponding clustered correlation matrix (right subplot) with five distinct clusters.

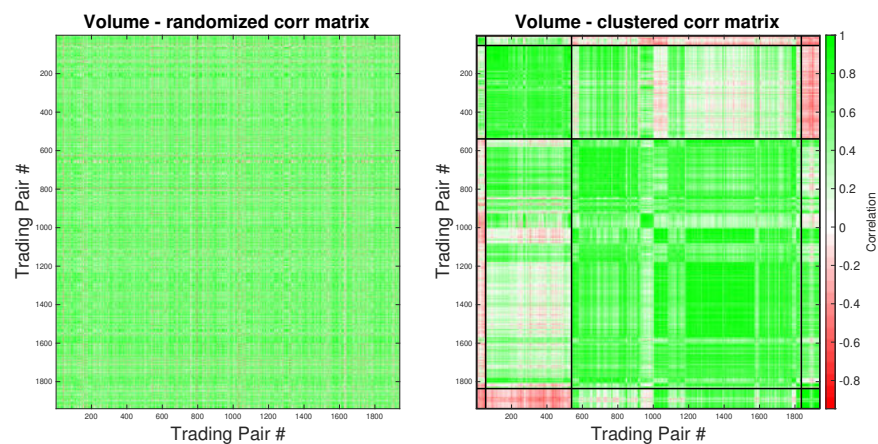


Figure 12: **Hierarchical clustering correlation matrix - Volume.** This Figure plots the randomized correlation matrix for the log trading volume (left subplot) and the corresponding clustered correlation matrix (right subplot) with five distinct clusters.

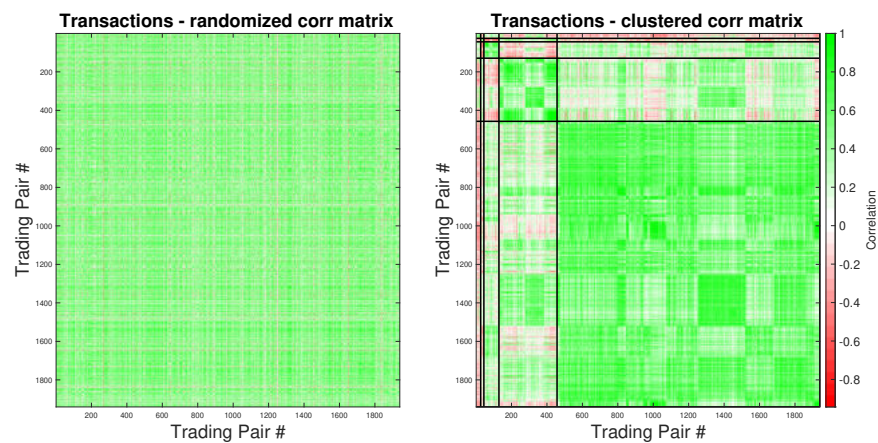


Figure 13: **Hierarchical clustering correlation matrix - Transactions.** This Figure plots the randomized correlation matrix for the number of transactions (left subplot) and the corresponding clustered correlation matrix (right subplot) with five distinct clusters.

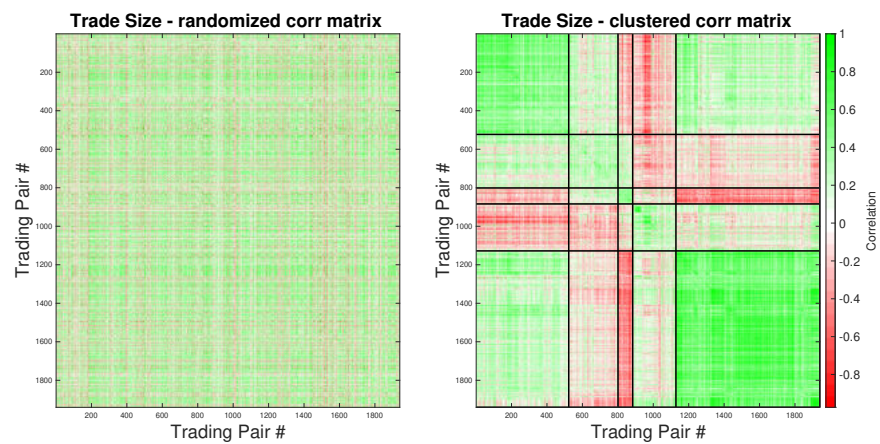


Figure 14: **Hierarchical clustering correlation matrix - Trade Size.** This Figure plots the randomized correlation matrix for the average trade size (left subplot) and the corresponding clustered correlation matrix (right subplot) with five distinct clusters.